

PCA Based Gene Selection

In this assignment, you have to implement a PCA based gene selection algorithm and show the list of the top selected genes as output. This document will guide you step by step.

Dataset

We are going to use gene expression arrays from the Gene Expression Omnibus (GEO) website. For example: [GSE6613](#) contains expression profiling by array of Parkinson's disease vs. controls. The data contains whole blood expression data from 50 patients with Parkinson's disease, 33 with neurodegenerative diseases other than PD, and 23 healthy controls. You can use the specified code or use your own libraries to load the data into a gene expression array/matrix.

Task

You have to perform gene selection based on PCA. Here is the algorithm:

1. Run **Principal Component Analysis (set variance = 0.95)**
2. Find the **loadings** of the top selected Principal Components
3. The **loadings will show coefficients** of loadings for each gene.
4. Select top genes having larger absolute value of the coefficients from the loadings.
 - a. Use top **5 PCAs** and provide weightage to the absolute values to the coefficients according to the variance they explain.
 - b. Add the coefficients multiplied by **variance explained** to get the ultimate weights of the genes
 - c. **Sort** the genes according to the values
5. Provide a **list of the top 20 genes** as output
6. Rerun the experiment using top 10 PCAs
7. Is there any difference between the two lists? Why is that happening? Comment on the suitability of this approach for gene selection. [put these answers as markup in your colab note book]

Code

The following notebook will help:

<https://colab.research.google.com/drive/1td7cL4Y499eUOCbqN0ov9wKChN6D9JLy?usp=sharing>

Rubric

You provided a runnable colab file, ensure you are reading the data using a library and your instructor is able to run the code without doing any changes	2.5 marks
Your solution is correct, uses loadings and sorts the genes according to weights	2.5 marks
Your solution shows correct output Two lists and differences in the two lists	2.5 marks
You have submitted timely	2.5 marks